ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #4

GisCraft v0.2.1

QGIS-class GIS in pure Rust: a clean-room 1:1 clone shell for QGIS muscle



v0.2.1	5 Crates	13460	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\giscraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

QGIS-class GIS in pure Rust: a clean-room 1:1 clone shell for QGIS muscle memory, with a modern core, CLI + MCP first, and no 2005 backend cruft.

What for: a workbench for asking questions of the world's free map (OpenStreetMap) and keeping the answers as your own map layers - scout locations, audit POIs, plan walks and hikes, verify what's really there.

Type a question, get a layer, analyze it, export it - by hand or via an AI agent (MCP). Full picture: docs/ECOSYSTEM.md.

What this wraps

QGIS (LTR 3.44), the free and open-source desktop GIS

- the reference our menus, docks, canvas behavior and shortcuts are cloned

from, black-box. No QGIS code ships here and none is needed to run. Full

picture: docs/WRAPPEE.md. Our plugin answer vs

plugins.qgis.org: docs/plugins.md.

Preview

!First light: 29 live Vienna pharmacies over OSM tiles

Rendered headless by the app itself (`--smoke` BMP preview). Streets: (c) OpenStreetMap contributors.

What You Can Do

How it runs: native egui app + CLI + MCP server over stdio, one shared command bus. SQLite project file *is* the project (our `.qgz`). Zero ports for the native track; no accounts, no keys (Photon/Overpass/tiles are keyless).

- Geocode places (Photon live, Vienna demo offline), search the map
- Query OpenStreetMap via fenced Overpass builder (dry-run QL or live execute)
- Ingest hits as layers (points, lines, polygons) into the project DB
- Render vectors over cached XYZ tiles: per-layer colors, concave fills, labels
- Pan/zoom-to-cursor/Fit/F5, layers dock with counts, menus + shortcuts
- Drive all of it from Claude via 9 MCP tools (same bus as the GUI)

Track B (reserved, not built): ports 11253/11254 are claimed for a future sandraschi webapp for users who never learned QGIS. The native app needs none.

Install

Pre-release: building from source is currently the only path (no installer,

no `.mcpb` yet). It takes one command chain:

```
winget install Rustlang.Rustup
# + "Desktop development with C++" (MSVC linker, required on Windows)
cargo run -p giscraft --
```

Everything else - Claude Desktop wiring, env vars, verification prompt -

is in INSTALL.md. Developers: docs/DEVELOPMENT.md.

Example Prompts

> "Search for Stephansplatz and show it on the map"

> *(geocode.search, demo offline or live with `--live`)*

> "Find all pharmacies in Vienna's 1st district and ingest them as a layer"

> *(overpass.query key=amenity value=pharmacy + bbox, `--live --ingest`)*

> "List the project layers with feature counts"

> *(layer.list - sources, geometry, visibility, counts)*

Documentation

Doc	Contents
Installation	All install methods, prerequisites
Onboarding	First run: QGIS side-by-side, keyless APIs, smoke test
Wrapped app	QGIS: what it is, official links, community
Plugin strategy	Why MCP tools are our plugin system
Architecture	L1/L2/L3, command bus, DB-as-project, stdio framing
Configuration	Env vars, config options
Tool Reference	All 10 MCP tools
Glossary	Plain-language terms: bbox, Overpass, tiles, CRS, ODbL...
Parity target	QGIS scope, oracle loop
UI parity	Measured checklist vs running QGIS
Development	Setup, recipes, gates
Troubleshooting	Common issues
Assessment	Tentative self-review + cross-checks

Requirements

Windows 10/11 64-bit (primary), Rust stable via rustup, MSVC linker

("Desktop development with C++"). No accounts, no API keys, no Docker.

License

MIT OR Apache-2.0. QGIS itself is GPLv2+ and not included; map tiles

(c) OpenStreetMap contributors (ODbL) - attribution shown in-app.

Status - does today / next week / non-goals

Does today: 5 crates + xtask; 25 docs files; 133 test fns; justfile; clone-shell GUI (QGIS muscle memory) + headless MCP/CLI (9 tools); `--smoke` BMP map preview; keyless Photon/Overpass with offline demo.

Next week: 3 democraft recipes (geocode-layer, overpass-fenced, render-preview); travelcraft boundary paragraph; live-service version/rate pinning.

Non-goals: not full QGIS parity (shell only, on purpose); web track reserved, not built.

Lineage - the Storytold craft-app pattern

Storytold ArtCraft (storytold/artcraft - "the IDE for artists") was the starting point. The 16 craft apps in the Storytold stable each ship one binary that is both the human CLI and the agent MCP server (stdio), headless-first, with `run.json` + report artifacts. That modus operandi and harness is what this repo implements; the code here is independent, the pattern is shared.

Complete MCP Tool & CLI Verb Reference

`mcp --connect PORT` drives the running app (`giscraft -- --control PORT`)
for the DB-touching tools, so there is a single writer. See
control-protocol.md. Without `--connect` everything
runs headless against the project DB file.

Every command is also a CLI subcommand (`--json` for agents). MCP transport:
stdio, MCP 2024-11-05, `initialize` / `tools/list` / `tools/call` / `ping`.

Tool	Required	Optional	What
`status`	-	-	App mode, command list, agent contract
`project.status`	-	-	Project CRS + layer count
`layer.add`	`id`, `name`	`source`, `geom_type`, `crs`, `visible`, `db`	Add (upsert) a layer, persists to SQLite
`layer.list`	-	`db`	Layers in draw order + `visible` + per-layer feature counts
`layer.import`	`path`	`format` (auto\	geojson\	csv), `layer_` `db`
`layer.remove`	`id`	`db`	Remove a layer and its features
`geocode.search`	`query`	`limit` 1..50, `live`	Photon demo (Stephansplatz) or live lookup
`overpass.query`	`key`, `value`	`south/west/north/east`, `live`, `ingest`, `db`	Fenced tag builder; dry-run QL by default; `live` executes (cap 100); `ingest` writes hits to DB (requires `live`)
`features.list`	-	`layer_id`, `db`	Ingested point features (cap 500), GeoJSON in `geom_json`
`processing.buffer`	`layer_id`	`distance` (meters, def 200), `segments` (8..256), `output_layer`, `db`	Great-circle buffer point features into circle polygons; persists to DB
`map.forge`	`path`	`live`, `ingest`, `db`	MapSpec JSON file -> offline QL plan per layer; `live` executes (cap 100/layer); `ingest` writes live hits to DB (requires `live`)
`wkt.show`	`wkt`	`layer_id`, `db`	WKT string (POINT/LINESTRING/POLYGON + MULTI, 2D only) as a layer; persists to DB

Tool	Required	Optional	What
`processing.grid`	`south`, `west`, `north`, `east`	`cell_size` (meters, def 500), `hex`, `output_layer`, `db`	Rect or hex grid over a bbox (cap 10k cells); persists to DB
`processing.nearest`	`source_layer`, `target_layer`	`output_layer`, `db`	Hub lines from each source feature to its closest target feature; persists to DB
`processing.geodesic`	`shape`, `lng`, `lat`	`radius_m` (circle), `major_m` / `minor_m` / `bearing_deg` (ellipse), `lng2` / `lat2` (line), `segments`, `output_layer`, `db`	True-distance circle/ellipse/line at a center point; persists to DB
`layer.color`	`id`	`color` (#rrggbb, empty clears to palette), `db`	Set a layer's display color; persists to DB
`doppler.locate`	`path`	`south` / `west` / `north` / `east` (required with path), `step_deg`, `top_n`, `terrain_m`, `output_layer`, `db`	Rank grid candidates by matching Doppler curves to ADS-B tracks; persists top hits to DB
`export.png`	-	`width`, `height`	Typed `unsupported` until the render path lands (Phase 1+)

CLI equivalents: `status`, `project-status`, `layer-add`, `layer-list`, `layer-import`, `layer-remove`, `geocode-search [--live]`, `overpass-query [--live] [--ingest] [--db]`, `features-list`, `processing-buffer`, `processing-grid --south --west --north --east [--cell-size] [--hex]`, `processing-nearest --source-layer --target-layer`, `processing-geodesic --shape circle --lng --lat --radius_m`, `layer-color --id --color`, `doppler-locate --path --south --west --north --east`, `forge-from-spec --path SPEC [--live] [--ingest] [--db]`, `wkt-show --wkt GEOM [--layer-id] [--db]`, plus `mcp` (serve, `--connect` for the live app). Per-command `--db` overrides `$GISCRAFT_DB`, default `./giscraft.db`. Forge depot `depot/maps/*/spec.json` checked by `cargo run -p xtask -- forge`.

Hostile input (empty ids, injection in tag keys, out-of-range limits/bbox) returns typed `invalid input` errors - never a panic, never silent.